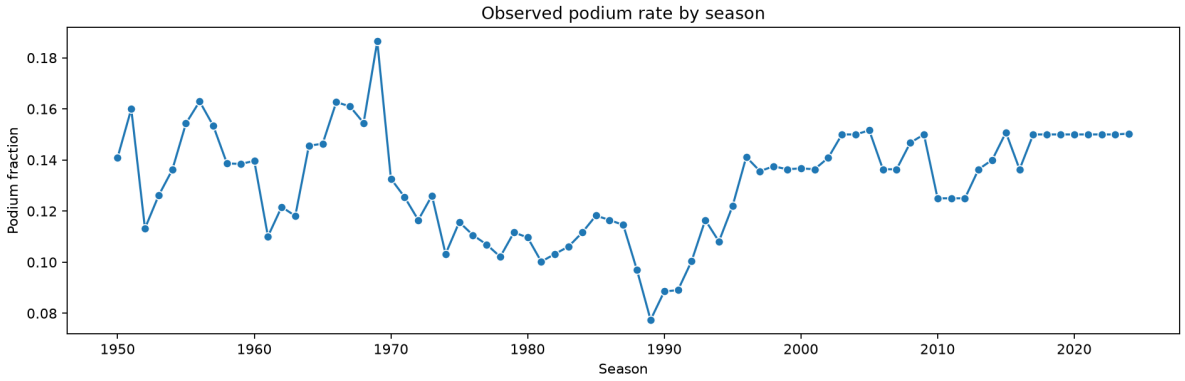
FORMULA 1
PODIUM PREDICTION

Companion | Milestone 1

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Reproducible historical analysis and pre-race prediction using the Kaggle Formula 1 World Championship data release.

ANALYSIS WINDOW	1950-2024
OBSERVATION GRAIN	One driver in one race
PREDICTION CUTOFF	After qualifying and final grid; before race start
MODEL SELECTION	Validation PR-AUC; threshold selected on validation
RUN DATE	16 October 2026



Prepared as an academic project companion. Results are descriptive and predictive; they do not establish causal effects.

Executive summary

This project audits and analyzes the historical Formula 1 tables, builds a leakage-aware entrant-level feature table, answers three data-engineering questions, and compares three model families. The final modeling table contains **26,668 driver-race records**; after deterministic duplicate resolution, each row represents one unique (raceId, driverId). The chronological split is 24,529 train rows, 780 validation rows, and 1,359 untouched test rows.

Validation selection: Shallow Neural Network has the highest validation PR-AUC (0.734); its threshold was selected using validation F1 (0.456). On the 2022-2024 test window it obtains PR-AUC 0.666, ROC-AUC 0.923, and F1 0.644. The Random Forest has the strongest test PR-AUC (0.676), despite lower validation PR-AUC, illustrating why the temporal holdout should remain untouched during selection.

The main modeling pattern is that recent grid and qualifying information carries strong signal, but historical out-of-time performance varies by model. A small home-versus-away rate difference remains after matching grid bins. Mechanical-retirement rates depend on a transparent but imperfect status-text mapping.

Project package: executed notebook, report, tables, figures, model artifact, and run instructions are included in the accompanying project archive. The source dataset archive is not duplicated; download it from the linked Kaggle source and point the notebook to its CSV folder.

1. Problem, target, and data

Prediction target. Podium is 1 when a driver finishes in positions 1-3 and 0 otherwise. The target uses post-race positionOrder only for label construction and historical description. Prediction time is immediately before race start, after qualifying and final-grid determination. Current-race points, position, status, laps, times, fastest lap, pit stops, and standings are prohibited model inputs.

Data source. Rohan Rao's Kaggle Formula 1 World Championship dataset, linked at the end of this report. The archive used in this run contains 14 Ergast-style CSV tables and covers seasons 1950-2024 (75 seasons). It includes race, result, qualifying, driver, constructor, circuit, status, standings, lap-time, pit-stop, sprint, and related tables. The notebook recursively discovers the CSV files and records their paths and shapes.

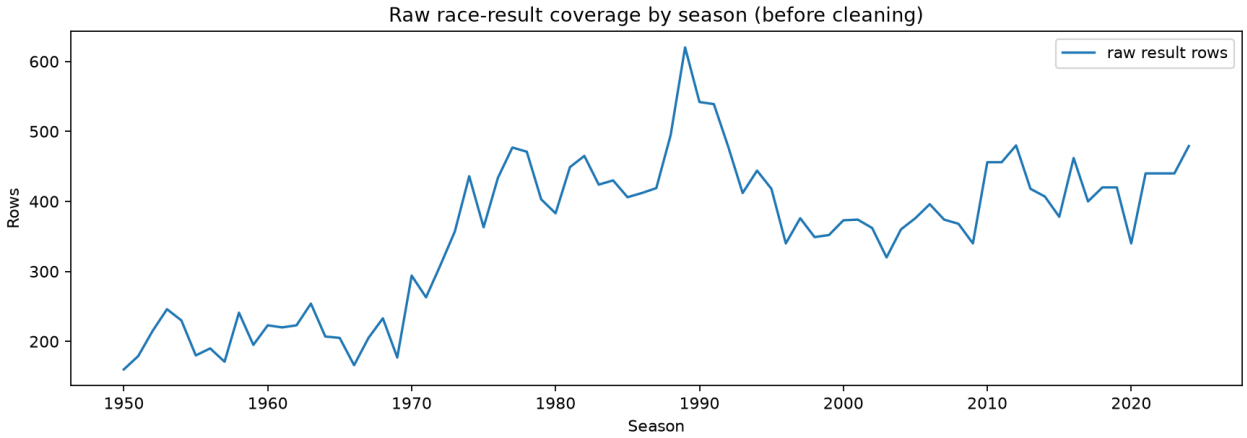
The project's analysis grain is one result entry for a specific driver and race. Raw result records are reduced from 26,759 to 26,668 unique race-driver rows after resolving 91 duplicated keys with a deterministic rule. The rule is documented in the cleaning log; the modeling table asserts unique race-driver keys.

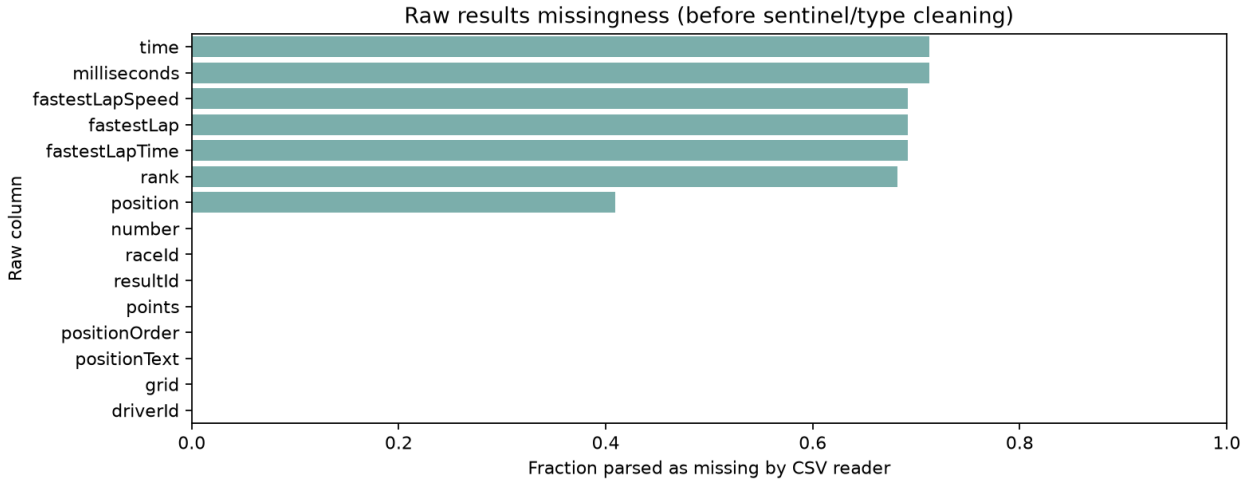
Table	Rows	Columns
circuits	77	9
constructors	212	5
constructor_results	12,625	5
constructor_standings	13,391	7
drivers	861	9
driver_standings	34,863	7
races	1,125	18
results	26,759	18
qualifying	10,494	9
lap_times	589,081	6
pit_stops	11,371	7
sprint_results	360	16
seasons	75	2
status	139	2

Key audits report no duplicate primary keys in the audited dimensions and no orphan foreign keys for the checked relationships. Missing fields are retained and handled according to their role rather than dropping all incomplete historical records. `grid=0` and unknown grid values are flagged; they are not treated as ordinary grid slots.

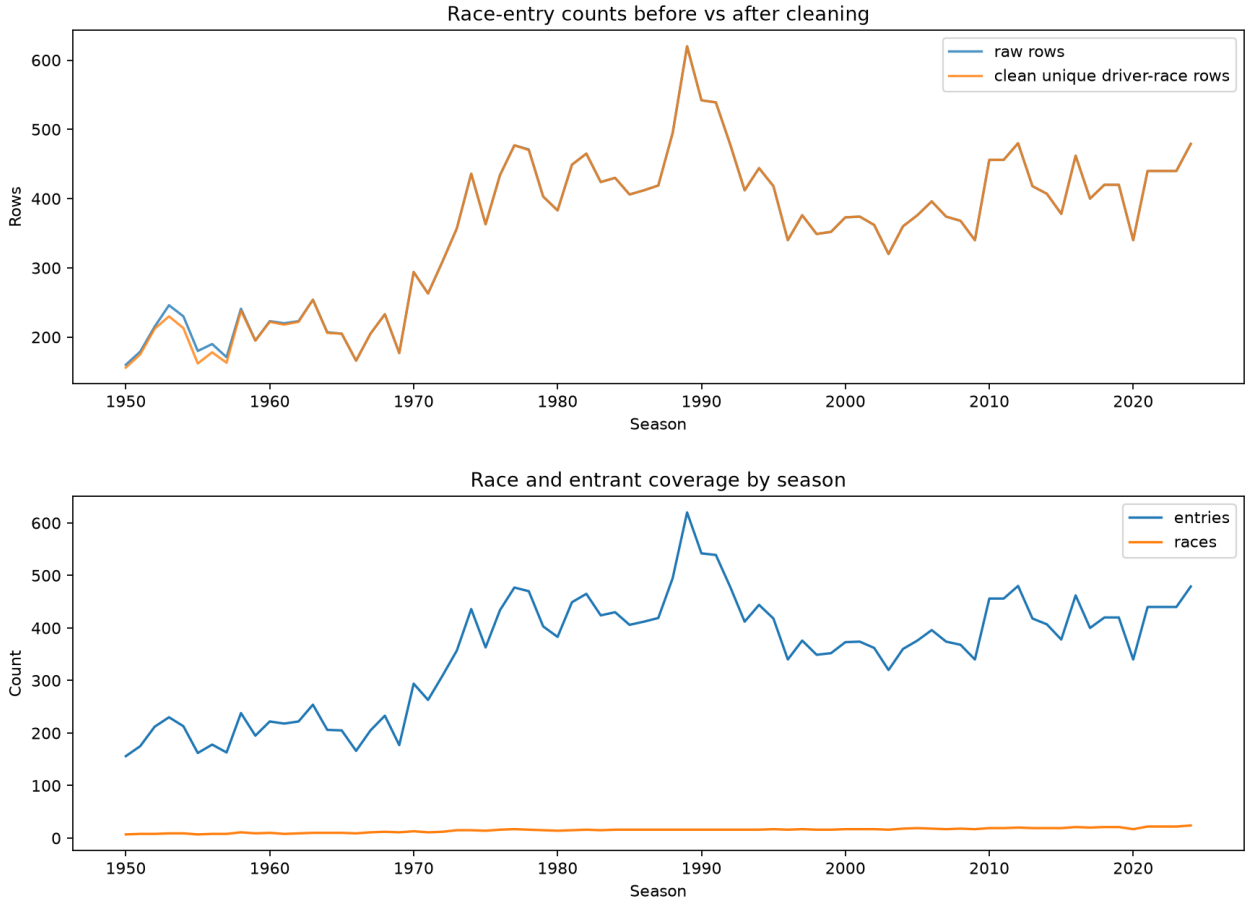
2. Exploratory analysis and data quality

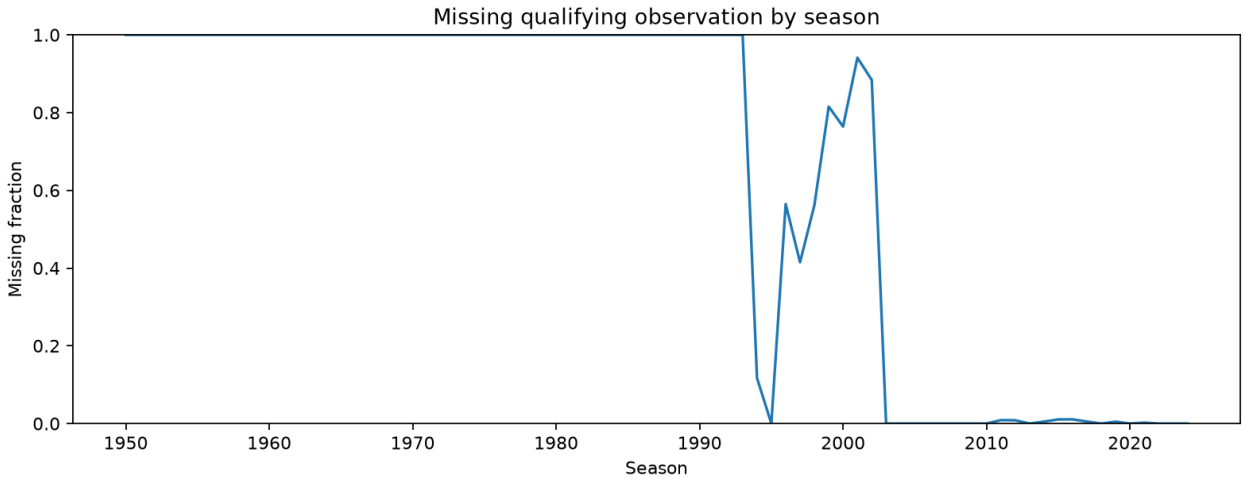
Before cleaning, the notebook plots raw result-row coverage, missingness, and duplicate race-driver groups. After cleaning, it compares raw and unique race-driver rows by season and plots the cleaned coverage and feature distributions. These paired views expose how duplicate-key resolution and missing data affect the analysis.

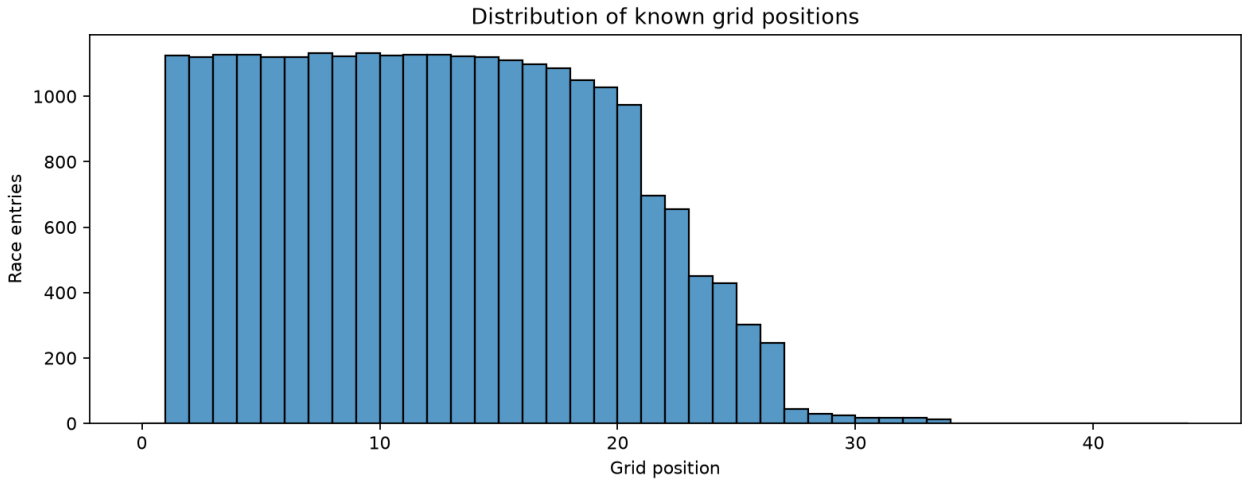


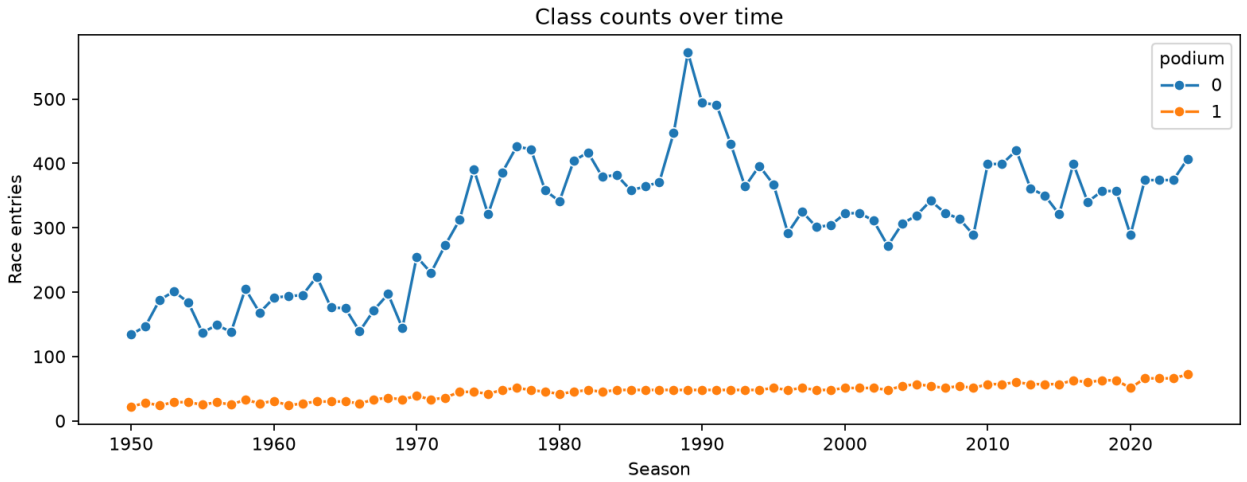


Coverage expands and changes over time: race counts, field sizes, qualifying availability, and rule eras differ across the 75-season span. The analysis therefore uses a chronological evaluation rather than a random split. The podium target is a minority outcome at entrant level, so PR-AUC is the primary ranking metric. The next figure compares raw and cleaned race-entry counts.

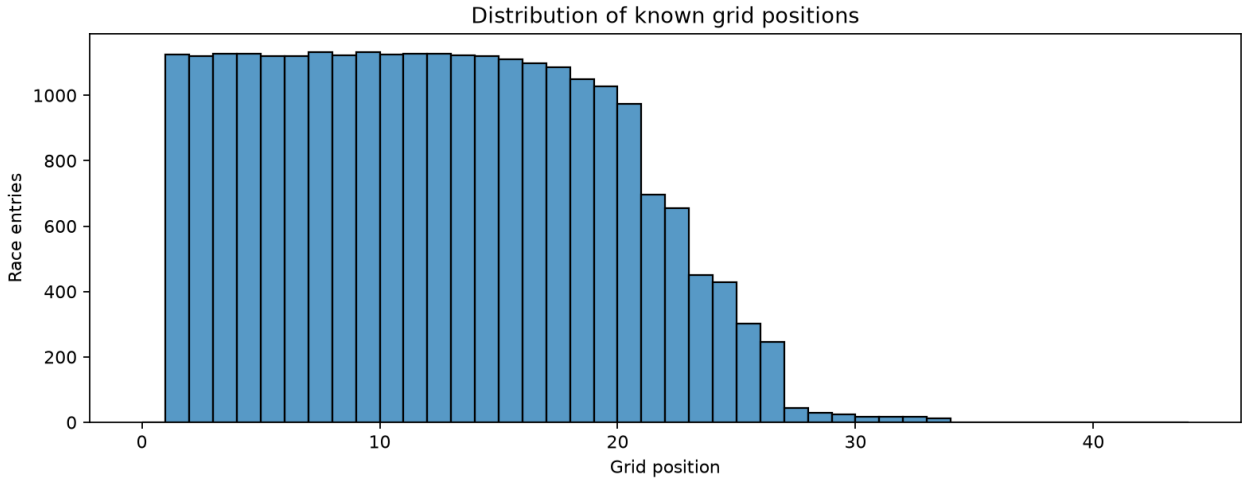


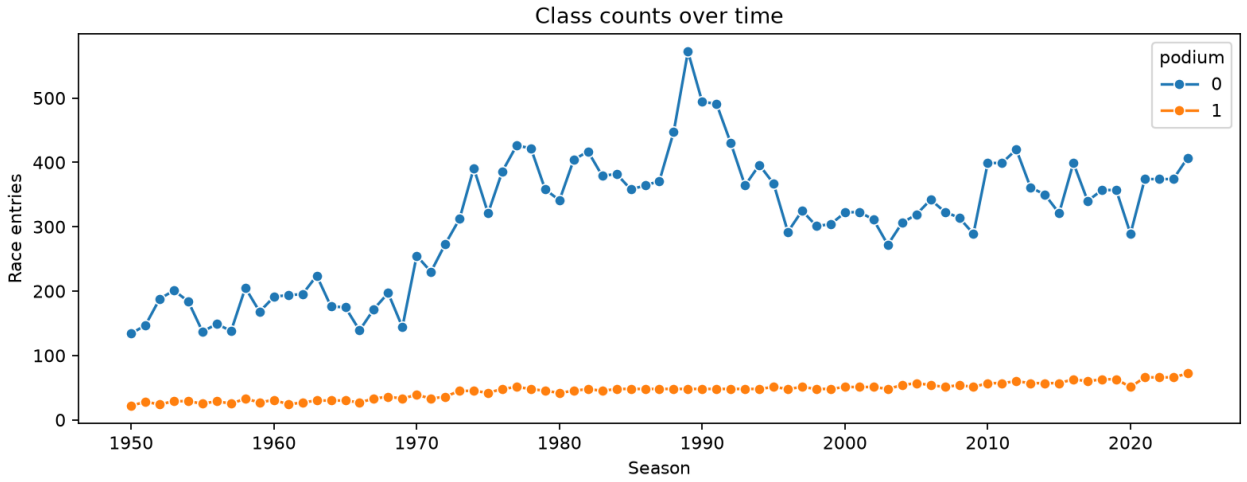

Missing qualifying observation by season
Missing fraction
Season
0.0
0.2
0.4
0.6
0.8
1.0
1950
1960
1970
1980
1990
2000
2010
2020


Distribution of known grid positions
Race entries
Grid position
0
200
400
600
800
1000
0
10
20
30
40


Class counts over time
Race entries
Season
podium
0
1
0
100
200
300
400
500
1950
1960
1970
1980
1990
2000
2010
2020

The entrant table preserves early historical rows even when qualifying observations are absent. Qualifying coverage is much higher in modern seasons; using it without missingness-aware preprocessing would make early-era predictions incomparable. The preprocessing pipeline imputes numeric values, includes missingness indicators, and one-hot encodes categoricals.



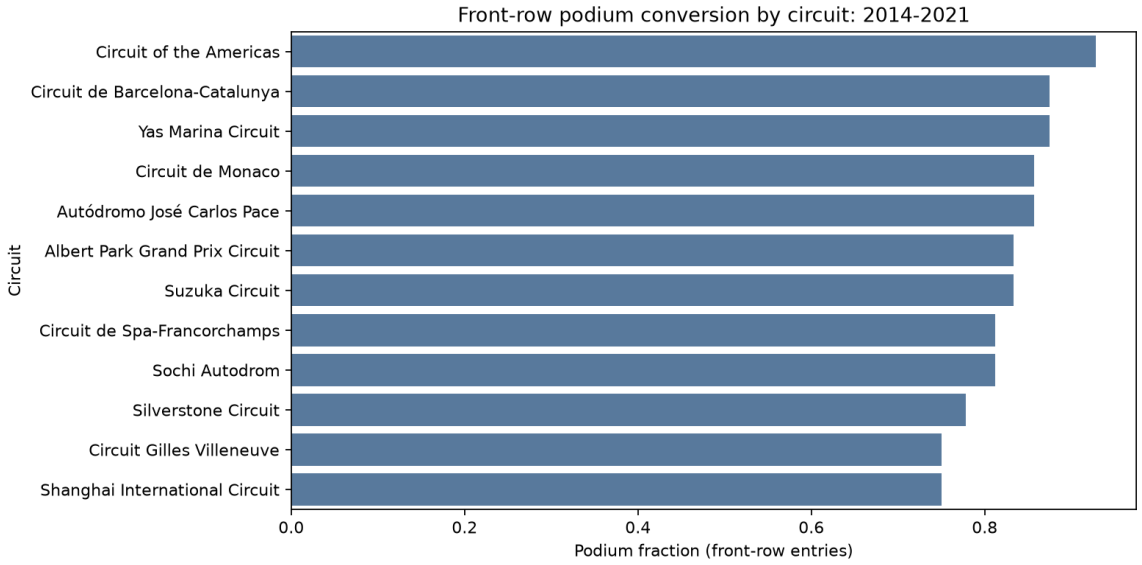


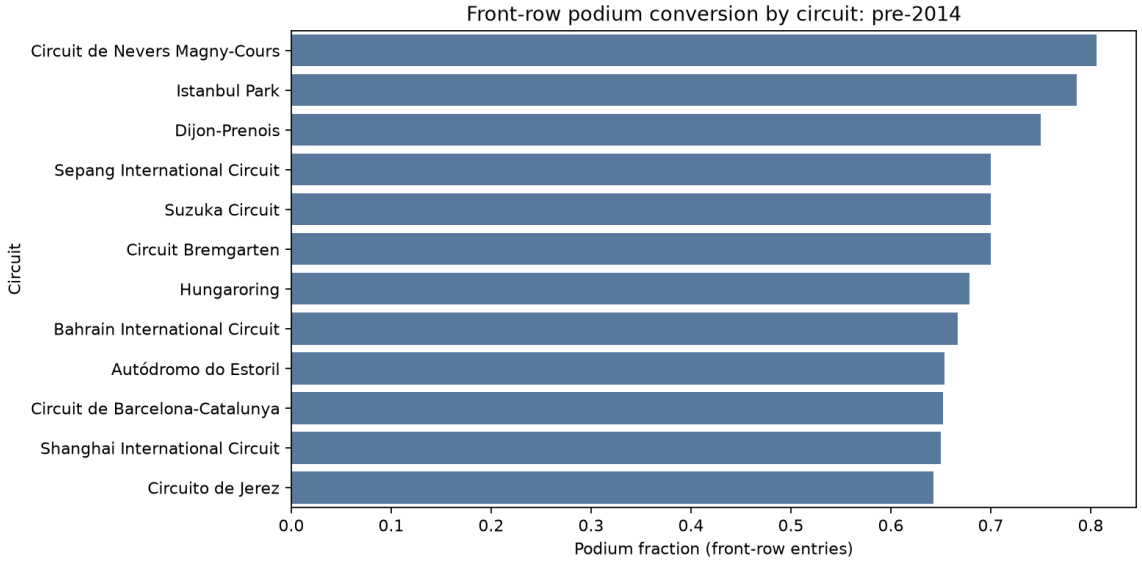
3. Data-engineering questions

3.1 Front-row conversion by circuit and era

Front row is defined by the **actual starting grid** positions 1 and 2, not qualifying rank. Penalties and pit-lane starts can make the two differ. The audit finds 2,509 differing positions among 10,494 entries with both values (23.9%); details are in `q1_grid_vs_qualifying_audit.csv`. Each circuit-period conversion rate is podium finishes divided by front-row starts. Rankings require at least 10 entries.

Finding: Circuit of the Americas leads eligible 2014-2021 circuits with 13 podiums from 14 front-row starts (92.86%). Barcelona-Catalunya and Yas Marina each record 14/16 (87.50%). Before 2014, Magny-Cours leads with 29/36 (80.56%). In 2022-2024, no circuit reaches the 10-entry threshold, so a stable recent-period ranking is unavailable. These are descriptive rates, not track effects.



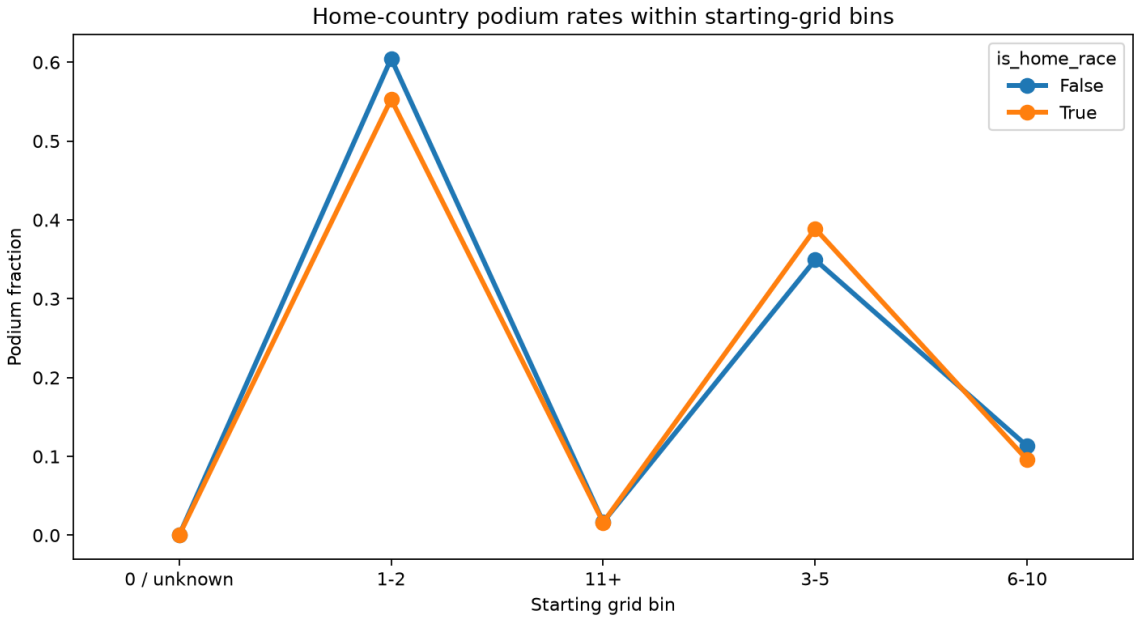


3.2 Home-country podium rates, compared at similar grid positions

Driver nationality and circuit country are normalized to a shared country vocabulary. Podium rates are compared within grid bins and standardized over bins represented in both home and away groups. This adjusts descriptively for starting position, not all driver, team, track, or era differences. The primary estimate retains all championship races; the Indianapolis 500 sensitivity excludes the 1950-1960 Indianapolis events, which had a distinct oval and entry context.

Finding: With all championship races, the standardized podium rate is 12.44% for home entries and 12.77% for away entries (difference -0.32 percentage points). Excluding the Indianapolis 500, the rates are 12.61% and 12.81% (difference -0.20 points). The difference remains small in both scopes. Country mismatches are listed for review; neither estimate is causal.

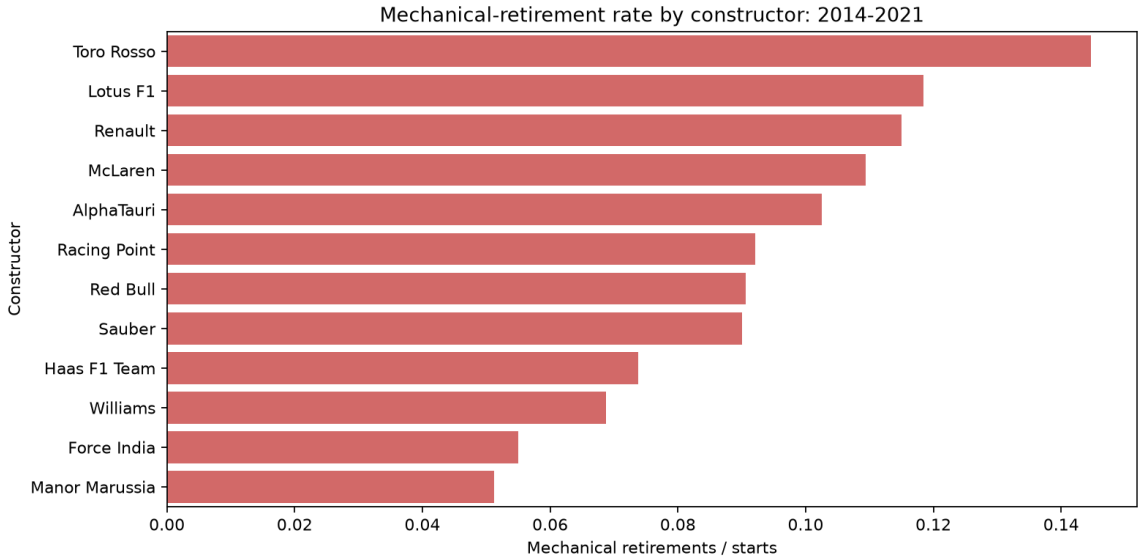
Q2 scope	Entries	Home rate	Away rate	Difference (pp)
all championship races	26,646	12.44%	12.77%	-0.32
excluding Indianapolis 500 (1950-1960)	26,265	12.61%	12.81%	-0.20



3.3 Constructor mechanical-retirement rates

Status text is mapped to mechanical categories using explicit keyword rules. The denominator is constructor starts; only constructors with at least 50 starts in a period are ranked. `q3_status_mapping.csv` preserves the source labels and assigned categories for audit.

Finding: In 2014-2021, Toro Rosso is highest among eligible constructors at 35 mechanical retirements from 242 starts (14.46%), followed by Lotus F1 at 9/76 (11.84%) and Renault at 23/200 (11.50%). In 2022-2024, Alpine is highest among eligible constructors at 13/136 (9.56%). Results depend on the mapping: labels such as accident, collision, engine, gearbox, and electrical can be ambiguous, and not every retirement cause is represented consistently.



4. Feature engineering and leakage controls

Features available by the prediction cutoff include final grid and grid flags, qualifying position and parsed session times, qualifying gaps, race-season context, driver and constructor nationality, home-race flag, field size, and prior driver/constructor form. Prior form features include counts, finish means, and podium rates over expanding history and recent windows. Driver/constructor histories are shifted by one race before rolling or expanding summaries are joined. Previous standings are shifted before use. Qualifying and lap-time duration strings are parsed to seconds while source strings remain preserved; lap-time observations are post-race and never enter model features. Sprint results are excluded because their timestamps are not guaranteed to precede the main race. Indianapolis 500 rows remain in the historical championship table, with an explicit Q2 sensitivity check.

A feature allowlist is used to form the model table, and assertions guard against post-race leakage. Numeric features are imputed and scaled for the linear and neural models; categorical variables are imputed and one-hot encoded with unseen-category handling. Transformations are fitted on training rows only. No validation/test resampling is performed. The full feature dictionary is supplied in `tables/feature_dictionary.csv`.

The selected time split trains on seasons through 2019, selects models and thresholds on 2020-2021, and reserves 2022-2024 for one final temporal evaluation. There are 17 validation races and 68 test races. The validation window is small, which adds uncertainty to model ranking and threshold selection.

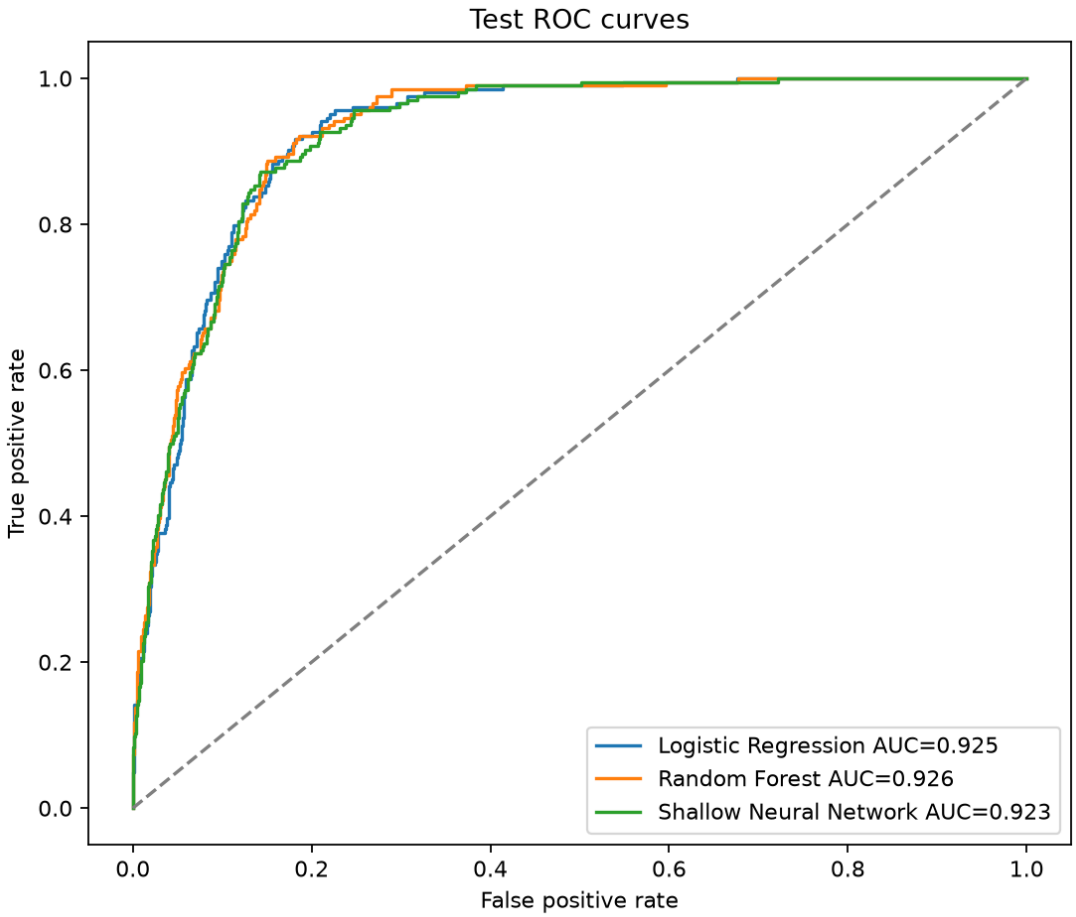
Partition	Seasons	Entrant rows
Train	1950-2019	24,529
Validation	2020-2021	780
Test	2022-2024	1,359

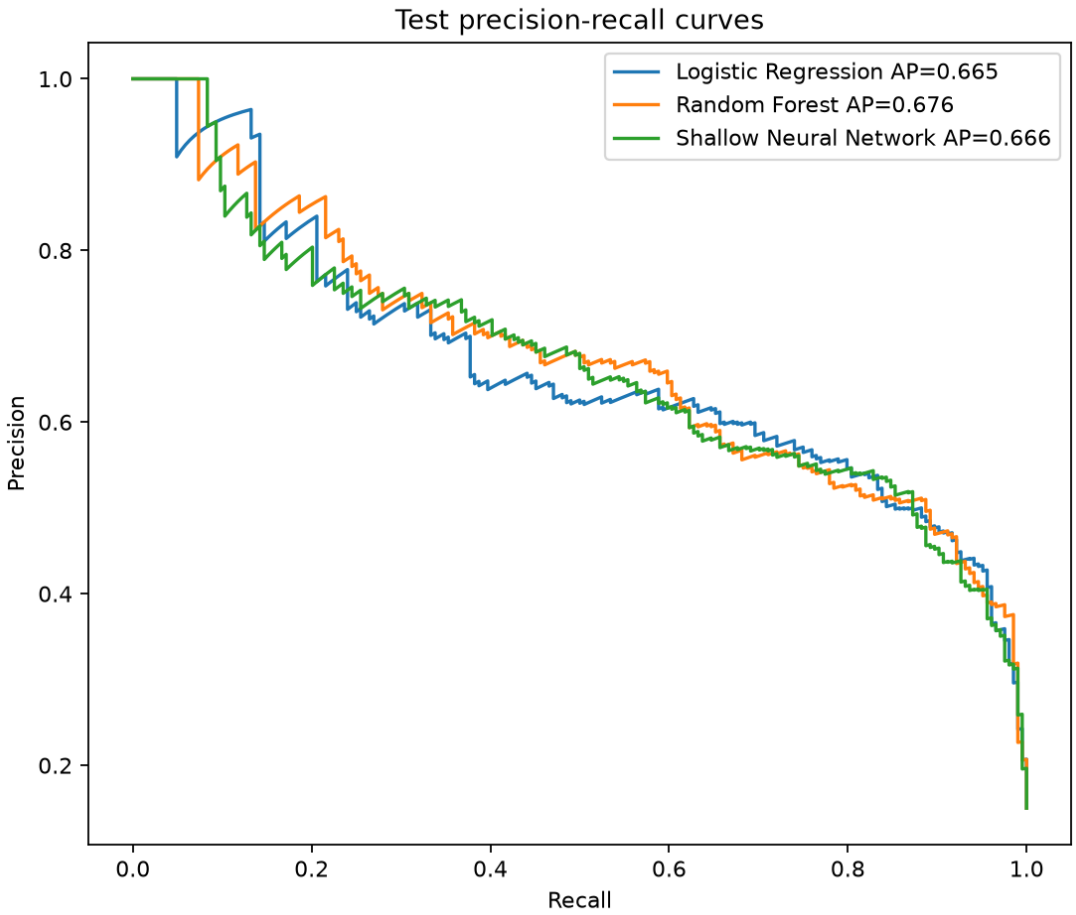
5. Models, tuning, and evaluation

Three model families were trained on the same rows, features, and chronological partitions: class-weighted Logistic Regression; a class-weighted Random Forest; and a shallow MLPClassifier with two hidden layers (64 and 32 ReLU units). TensorFlow was unavailable in the verified environment, so the scikit-learn neural-network fallback was used. It trains one epoch at a time, monitors external validation log loss, stops after 12 epochs without improvement, and restores the best validation-loss weights (epoch 5). The plots show training/validation loss and validation PR-AUC/ROC-AUC by epoch. F1 thresholds are selected on validation; validation PR-AUC selects the model; test data are not used for tuning. PR-AUC is primary for this imbalanced target, with F1 as the secondary thresholded metric.

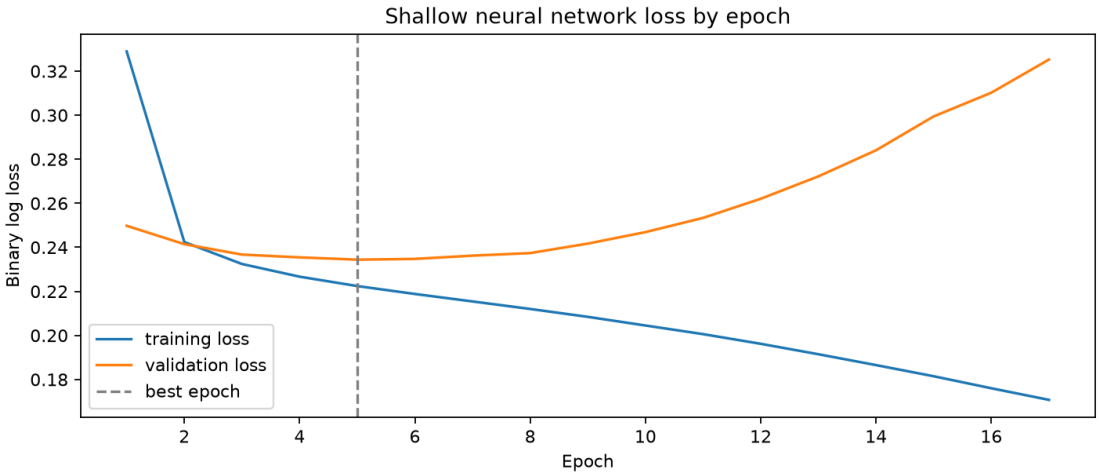
Model	Split	PR-AUC	ROC-AUC	F1	Precision	Recall
Logistic Regression	train	0.573	0.918	0.567	0.563	0.572
Logistic Regression	validation	0.680	0.923	0.701	0.711	0.692
Logistic Regression	test	0.665	0.925	0.647	0.574	0.740
Random Forest	train	0.630	0.920	0.585	0.589	0.580
Random Forest	validation	0.697	0.918	0.704	0.717	0.692
Random Forest	test	0.676	0.926	0.615	0.608	0.623
Shallow Neural Network	train	0.618	0.925	0.560	0.629	0.505
Shallow Neural Network	validation	0.734	0.924	0.700	0.683	0.718
Shallow Neural Network	test	0.666	0.923	0.644	0.544	0.789

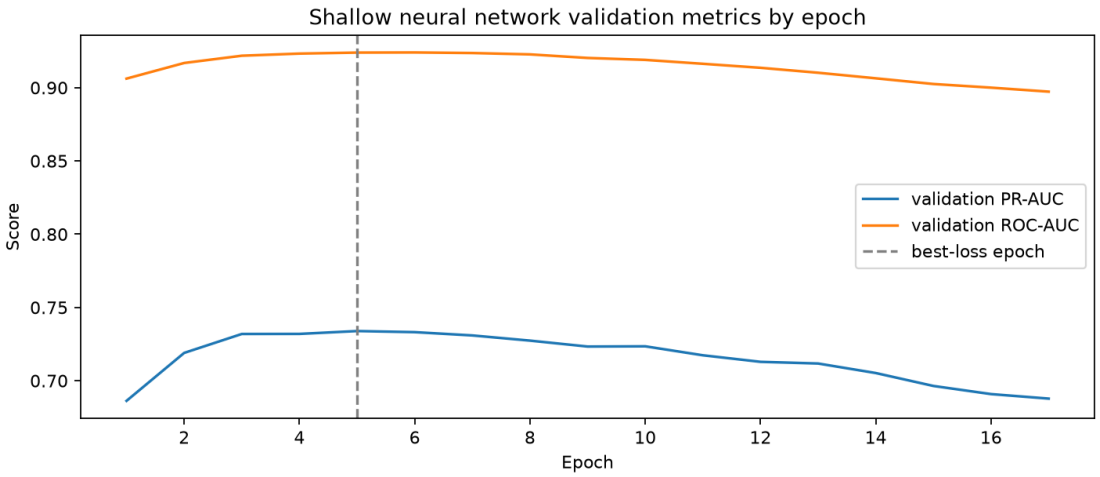
Validation-selected model: Shallow Neural Network; validation PR-AUC=0.734, ROC-AUC=0.924, F1=0.700; frozen threshold=0.456. The validation-selected MLP test PR-AUC is 0.666, ROC-AUC=0.923, F1=0.644. The Random Forest test PR-AUC is 0.676; the fact it leads on the test does not justify replacing the precommitted validation selection after seeing the test. This gap is evidence of temporal uncertainty and modest sample size.


Test ROC curves
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression AUC=0.925
Random Forest AUC=0.926
Shallow Neural Network AUC=0.923


Test precision-recall curves
Logistic Regression AP=0.665
Random Forest AP=0.676
Shallow Neural Network AP=0.666
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0

Neural-network learning diagnostics

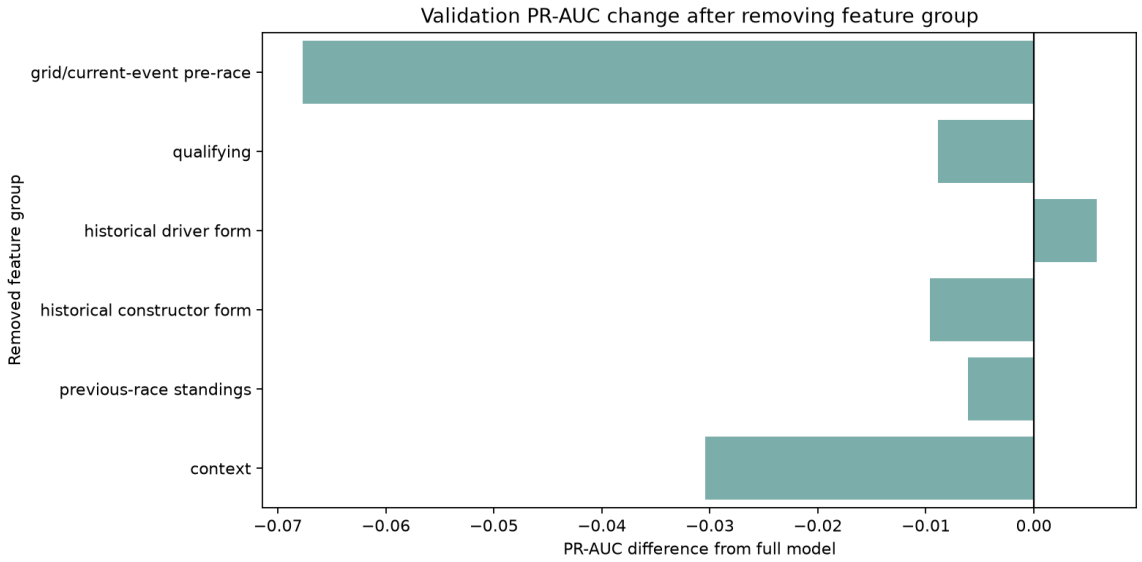




6. Ablation and preprocessing experiments

Ablations remove one feature group at a time and re-fit the selected model family using training rows; each threshold is again selected on validation. Deltas below compare validation PR-AUC with the full selected model (0.734). These are measures of predictive reliance under this dataset and pipeline, not causal effects.

Removed group	Features	Validation PR-AUC	Delta
grid/current-event pre-race	3	0.666	-0.068
qualifying	5	0.725	-0.009
historical driver form	5	0.740	0.006
historical constructor form	5	0.724	-0.010
previous-race standings	4	0.728	-0.006
context	11	0.703	-0.030



Removing the grid/current-event pre-race group causes the largest validation PR-AUC drop (-0.068); removing qualifying features causes a smaller drop (-0.009). Removing prior driver form slightly improves PR-AUC (+0.006) in this particular validation window. This can reflect noise, sample variation, or redundancy and does not prove historical form is useless.

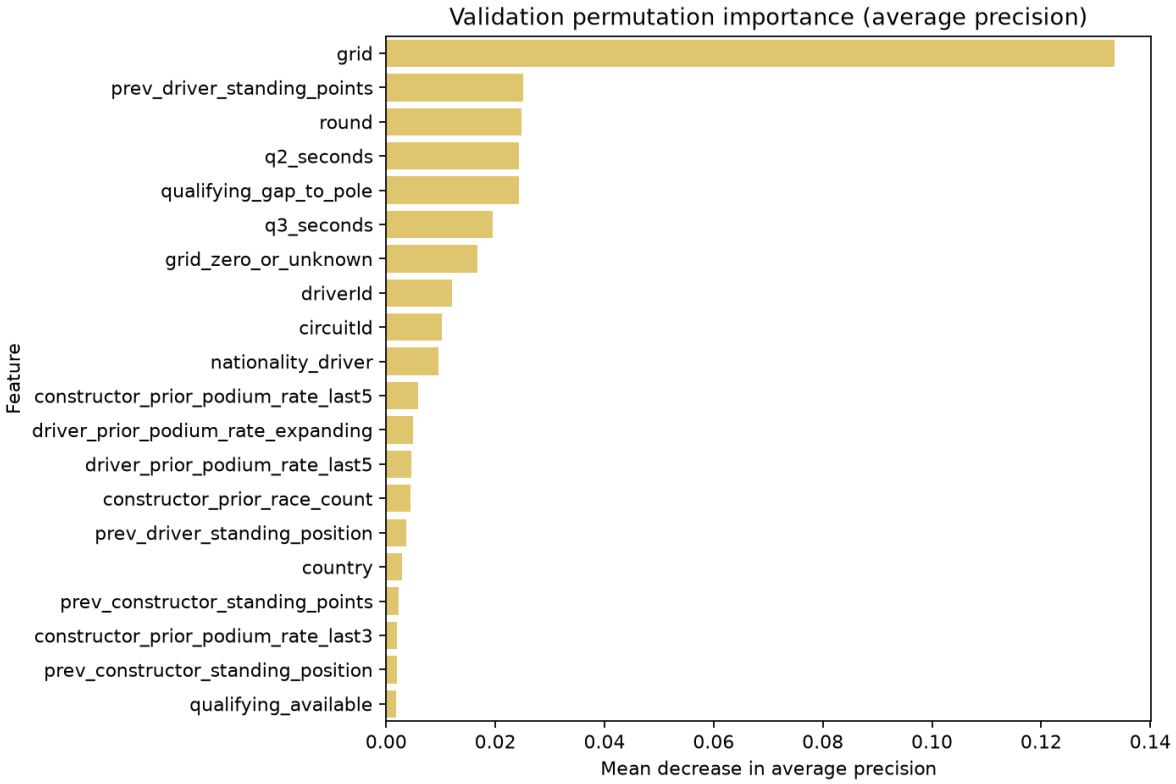
A controlled Logistic Regression experiment compares missingness indicators plus numeric scaling against a recipe with neither, holding the split and raw feature groups fixed. The no-indicator/no-scaling variant reaches validation PR-AUC 0.733 compared with 0.680. Since two choices change together, the difference cannot be attributed to either choice alone. The no-scaling variant also emitted a convergence warning at 1,500 iterations, so treat this result as tentative. The main three-family comparison retains its baseline preprocessing for consistency.

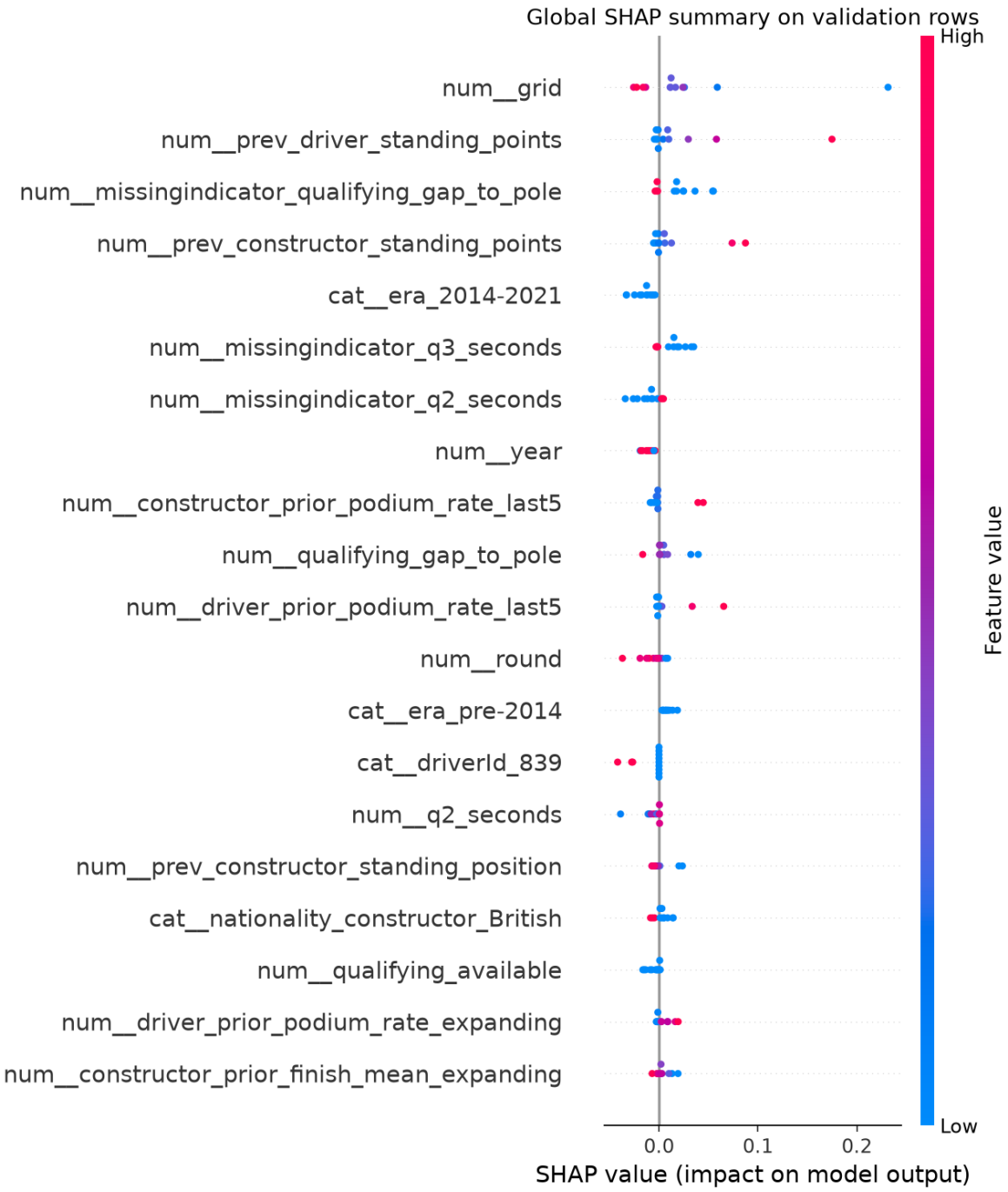
Preprocessing order	Validation PR-AUC	ROC-AUC	F1
impute-indicator-scale-encode	0.680	0.923	0.701
impute-no-indicator-no-scale-encode	0.733	0.924	0.710

7. Explainability

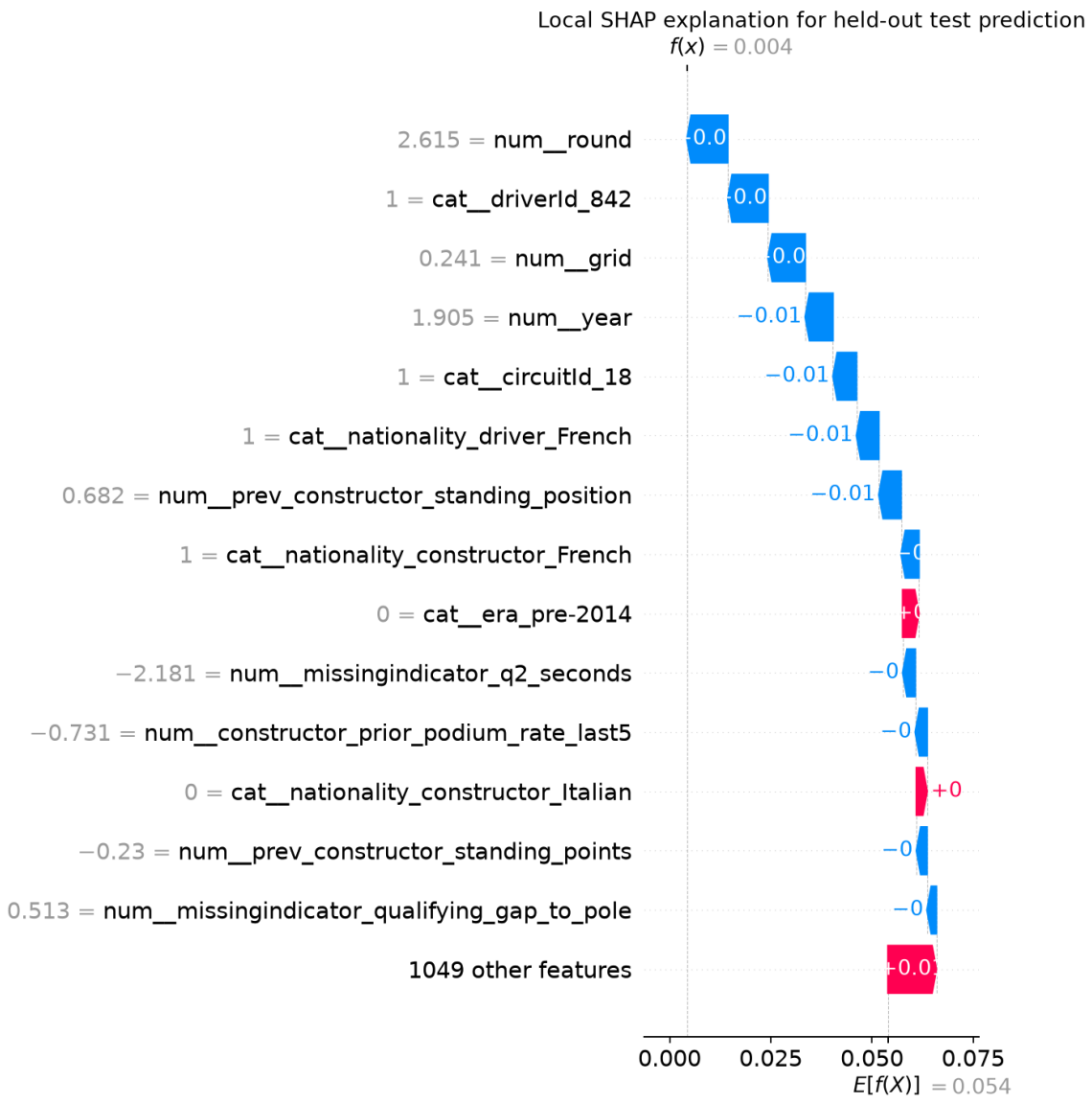
Global permutation importance uses held-out validation rows and reports mean decrease in average precision after permuting each original feature. SHAP explanations use the validation-fitted selected model and transformed model inputs; global SHAP summarizes a sample of validation rows, and local SHAP explains one held-out test prediction. Since the selected estimator is an MLP, the SHAP permutation explainer is used on a bounded sample. Transformed one-hot or missingness features may appear in the SHAP charts; they map back to the source variables listed in the feature dictionary.

Permutation feature	Mean AP decrease	Global SHAP feature	Mean |SHAP|
grid	0.133	num__grid	0.039
prev_driver_standing_points	0.025	num__prev_driver_standing_points	0.025
round	0.025	num__missingindicator_qualifying_gap_to_pole	0.021
q2_seconds	0.024	num__prev_constructor_standing_points	0.017
qualifying_gap_to_pole	0.024	cat__era_2014-2021	0.016
q3_seconds	0.020	num__missingindicator_q3_seconds	0.015
grid_zero_or_unknown	0.017	num__missingindicator_q2_seconds	0.012
driverId	0.012	num__year	0.011
circuitId	0.010	num__constructor_prior_podium_rate_last5	0.010
nationality_driver	0.010	num__qualifying_gap_to_pole	0.010





Local test case. The notebook selects the held-out row with the largest absolute probability-label error. Its row index is shown in notebook; the actual podium label is reported in notebook, predicted probability 0.0000, frozen threshold 0.0000, and predicted class reported in notebook. This is an intentionally difficult case (a missed podium if the actual label is 1), not a representative performance example.



Permutation importance and SHAP values explain model behavior from associations in the observed data. They do not show that changing a feature would cause a podium result. Correlated features, historical shifts, and small samples can redistribute or destabilize attribution.

8. Inference demonstration

The notebook provides `predict_podium_probability(record)`. It requires grid position, accepts missing optional pre-race fields, warns when they are imputed, handles unseen categorical values through one-hot encoding, and rejects recognized post-race fields. The selected model and validation-frozen threshold are returned with probability and class. Example outputs from the executed notebook:

Example	Model	Probability	Threshold	Class	Returned warning
Complete pre-race record	Shallow Neural Network	0.4307	0.4559	0	None
Partial fields + unseen category	Shallow Neural Network	0.5034	0.4559	1	Optional predictors imputed; unseen category accepted

The incomplete example intentionally keeps only required grid information and adds an unseen category; the function reports omitted predictors that were imputed. For a real pre-race production use case, historical form and qualifying fields should be assembled from data available at the same cutoff, and model calibration should be rechecked on the target season.

9. Limitations and interpretation

• The most recent test window is only three seasons, and the validation window is two seasons; year-to-year race and rule changes can shift the rankings.

• Qualifying coverage and the meaning of grid positions differ across historical eras. Missingness is not random and can encode era.

• The Indianapolis 500 (1950-1960) is retained for championship completeness; its oval and entry format can affect historical comparisons, so the home-race result includes an exclusion sensitivity.

• Only 17 races fall in validation and 68 in test; the number of podium outcomes is therefore modest. Thresholds and top-ranked circuit rates have sampling uncertainty.

• The dataset includes duplicate result records for 91 race-driver keys; deterministic resolution is logged, but alternative duplicate policies could affect edge cases.

• Home-country matching relies on imperfect nationality/country normalization. The comparison is descriptive and does not control for all confounders.

• Mechanical status labels are text-coded and ambiguous. The mapped rate is not a verified engineering failure rate.

• Driver and constructor identities can change or be unseen. One-hot models cannot infer reliable identity-specific effects for new entrants.

• The neural model ranked best by validation PR-AUC but did not lead on test PR-AUC. This shows selection uncertainty; test performance must be reported as observed.

• Ablations and explainability scores describe predictive associations, not causal effects. Model calibration and uncertainty intervals are not included in this milestone.

Reproducibility: the notebook fixes random seeds, records the discovered dataset and table audit, derives all report values from saved outputs, and exports the selected estimator plus frozen threshold. Run all notebook cells after pointing `F1_DATA_ROOT` to the extracted dataset directory.

10. Conclusion

The project builds a reproducible entrant-level Formula 1 analysis and completes the required three-family modeling comparison. Validation PR-AUC selects the shallow neural network at 0.734; on the untouched 2022-2024 test window it reaches PR-AUC 0.666, ROC-AUC 0.923, and F1 0.644. The Random Forest's stronger test PR-AUC (0.676) emphasizes that the validation ranking is uncertain, rather than changing the selection rule after test inspection.

The descriptive questions show high front-row conversion for selected circuit-period samples, a small negative home-versus-away difference after grid-bin matching, and constructor mechanical-retirement rates that vary by period under the stated status mapping. Each result is bounded by sample size, era differences, and source-label limitations. The model is best treated as a reproducible historical baseline rather than a race-weekend forecasting product.

Data citation

Rao, Rohan. Formula 1 World Championship (1950-2020), Kaggle dataset. The archive used here contains additional seasons through 2024. https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

Project evidence files: executed notebook; `tables/model_evaluation.csv`; `tables/ablation_results.csv`; `tables/preprocessing_experiments.csv`; `tables/feature_dictionary.csv`; and the figures embedded throughout this report.